

UNIT-I

HEAT TRANSFER BY CONDUCTION

1. What is conduction?

Heat conduction is a mechanism of heat transfer from a region of high temperature to a region of low temperature within a medium (solid, liquid or gases) or different medium in direct physical contact.

In conduction, energy exchange takes place by the kinematic motion or direct impact of molecules. Pure conduction is found only in solids.

2. Define convection.

Convection is a process of heat transfer that will occur between a solid surface and a fluid medium when they are at different temperatures. Convection is possible only in the presence of fluid medium.

3. Define radiation.

The heat transfer from one body to another without any transmitting medium is known as radiation. It is an electromagnetic wave phenomenon.

4. State Fourier's law of conduction.

The rate of heat conduction is proportional to the area measured normal to the direction of heat flow and to the temperature gradient in that direction.

$$Q \propto A \frac{dT}{dx}$$
$$Q = -kA \frac{dT}{dx}$$

Where, A- Area in m²

$$\frac{dT}{dx} - \text{Temperature gradient, } K / m$$
$$k - \text{Thermal conductivity, } W / mK$$

5. Write down the three dimensional heat conduction equations in Cartesian co-ordinate system.

The general three dimensional heat conduction equation in Cartesian co-ordinate is

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

Where

q- Heat generator – W /m²

α -Thermal diffusivity –m²/s

6. Write down the three dimensional heat conduction equations in cylindrical co-ordinate system.

The general three dimensional heat conduction equations in cylindrical co-ordinate is

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q}{K} = \frac{1}{\alpha} \frac{\partial T}{\partial \theta}$$

7. List down the three types of boundary conditions.

1. Prescribed temperature
2. Prescribed heat flux
3. Convection boundary conditions

8. Write down the equation for conduction of heat through a slab or plane wall

$$\text{Heat transfer, } Q = \frac{\Delta T_{\text{Overall}}}{R}$$

Where

$$\Delta T = T_1 - T_2$$

$$R = \frac{L}{kA} - \text{Thermal resistance of slab}$$

L – Thickness of slab

K – Thermal conductivity of slab

A - Area

9. Write down the equation for conduction of heat through a hollow cylinder.

$$\text{Heat transfer, } Q = \frac{\Delta T_{\text{Overall}}}{R}$$

$$R = \frac{1}{2\pi L K} \ln \left(\frac{r_2}{r_1} \right) - \text{Thermal resistance of slab}$$

L – Length of cylinder

K – Thermal conductivity

r_2 – outer radius

r_1 – Inner radius

10. Write down equation for conduction of heat through hollow sphere

$$\text{Heat transfer, } Q = \frac{\Delta T_{\text{Overall}}}{R}$$

$$\Delta T = T_1 - T_2$$

$$R = \frac{r_2 - r_1}{4\pi k(r_1 r_2)} - \text{Thermal resistance of hollow sphere}$$

11. State Newton's law of cooling or convection law.

Heat transfer by convection is given by Newton's law of cooling

$$Q = hA(T_s - T_\infty)$$

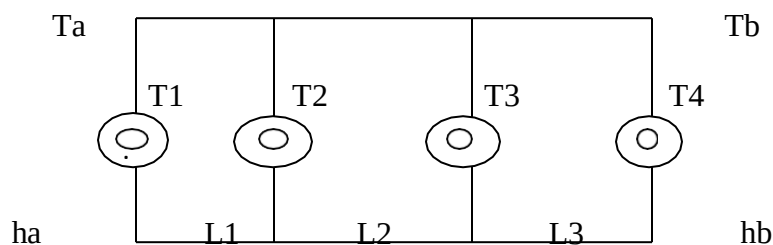
Where A – Area exposed to heat transfer in m^2

h – heat transfer coefficient in $\text{W}/\text{m}^2\text{K}$

T_s – Temperature of the surface in K

T_∞ – Temperature of the fluid in K

12. Write down the equation for heat transfer through a composite plane wall.



$$\text{Heat transfer, } Q = \frac{\Delta T_{\text{Overall}}}{R}$$

Where

$$\Delta T = T_a - T_b$$

$$R = \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}$$

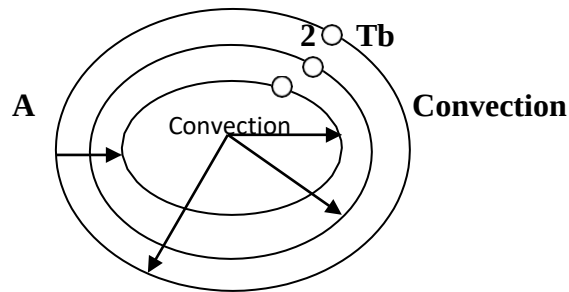
L – Thickness of slab

A – Area

h_a – Heat transfer coefficient at inner diameter

h_b – Heat transfer coefficient at outer side

13. Write down the equation for heat transfer through composite pipes or cylinder.



$$\text{Heat transfer, } Q = \frac{\Delta T_{\text{Overall}}}{R}$$

Where

$$\Delta T = T_a - T_b$$

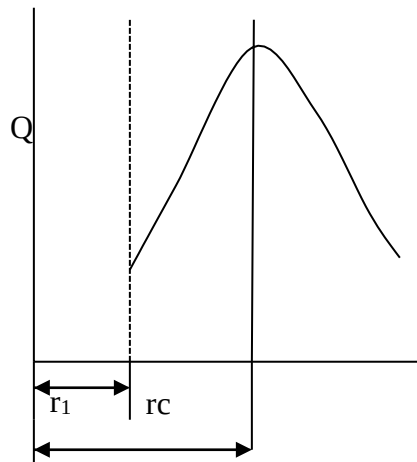
$$R = \frac{1}{2\pi L h_a r_1} + \frac{\ln\left(\frac{r_2}{r_1}\right)}{k_1} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{k_2} + \frac{1}{h_b r_3}$$

14. Define overall heat transfer co-efficient.

The overall heat transfer by combined modes is usually expressed in terms of an overall conductance or overall heat transfer co-efficient 'U'.

$$\text{Heat transfer, } Q = UA \Delta T.$$

15. What is critical radius of insulation or critical thickness?



Critical radius = r_c

Critical thickness = $r_c - r_1$

r_c = Critical radius

Addition of insulating material on a surface does not reduce the amount of heat transfer rate always. Infact under certain of circumstance it actually increases the heat loss up to certain thickness of insulation. The radius of insulation for which the heat transfer is maximum is called critical radius of insulation, and the corresponding thickness is called critical thickness.

16. Define fins or extended surfaces.

It is possible to increase the heat transfer rate by increasing the surface of heat transfer. The surfaces used for increasing heat transfer are called extended surfaces or sometimes known as fins.

17. State the applications of fins.

The main application of fins is

1. Cooling of electronic components
2. Cooling of motor cycle engines.
3. Cooling of transformers
4. Cooling of small capacity compressors.

18. Define Fin efficiency.

The efficiency of a fin is defined as the ratio of actual heat transferred to the maximum possible heat transferred by the fin

$$\eta_{fin} = \frac{Q_{fin}}{Q_{max}}$$

19. Define Fin effectiveness.

Fin effectiveness is the ratio of heat transfer with fin to that without fin

$$\text{Fin effectiveness} = \frac{Q_{\text{withfin}}}{Q_{\text{withoutfin}}}$$

20. What is meant by steady state heat conduction?

If the temperature of a body does not vary with time, it is said to be in a steady state and that type of conduction is known as steady state heat conduction.

21. What is meant by Transient heat conduction or unsteady state conduction?

If the temperature of a body varies with time, it is said to be in a transient state and that type of conduction is known as transient heat conduction or unsteady state conduction.

22. What is Periodic heat flow?

In periodic heat flow, the temperature varies on a regular basis.

Examples:

1. Cylinder of an IC engine.
2. Surface of earth during a period of 24 hours.

23. What is non periodic heat flow?

In non periodic heat flow, the temperature at any point within the system varies non linearly with time.

1. Heating of an ingot in a furnace.
2. Cooling of bars.

24. What is meant by Semi – infinite solids?

In a semi infinite solid, at any instant of time, there is always a point where the effect of heating or cooling at one of its boundaries is not felt at all. At this point the temperature remains unchanged. In semi infinite solids, the biot number value is ∞ .

25. What is meant by infinite solid?

A solid which extends itself infinitely in all directions of space is known as infinite solid.

In infinite solids, the biot number value is in between 0.1 and 100.

$$0.1 < Bi < 100.$$

26. Define Biot number.

It is defined as the ratio of internal conductive resistance to the surface convective resistance.

$$B_i = \frac{\text{Internal conductive resistance}}{\text{Surface convective resistance}}$$

$$B_i = \frac{hL_c}{k}$$

27. What is the significance of Biot Number.

Biot number is used to find Lumped heat analysis, Semi infinite solids and infinite solids

If $Bi < 0.1 \rightarrow$ Lumped heat analysis

$Bi = \infty \rightarrow$ Semi infinite solids

$0.1 < Bi < 100 \rightarrow$ Infinite solids.

28. Explain the significance of Fourier number.

It is defined as the ratio of characteristic body dimension to temperature wave penetration depth in time.

$$\text{Fourier Number} = \frac{\text{Characteristic body dimension}}{\text{Temperature wave penetration depth in time.}}$$

It signifies the degree of penetration of heating or cooling effect of a solid.

29. What are the factors affecting the thermal conductivity?

1. Moisture
2. Density of material
3. Pressure
4. Temperature
5. Structure of material.

30. What are Heisler charts?

In Heisler chart, the solutions for temperature distributions and heat flows in plane walls, long cylinders and spheres with finite internal and surface resistance are presented. Heisler charts are nothing but analytical solutions in the form of graphs.

UNIT-II

HEAT TRANSFER BY CONVECTION AND PHASE CHANGE

1. What is dimensional analysis?

Dimensional analysis is a mathematical method which makes use of the study of the dimensions for solving several engineering problems. This method can be applied to all types of fluid resistances, heat flow problems in fluid mechanics and thermodynamics.

2. State Buckingham Π theorem.

Buckingham Π theorem states as follows: "If there are **n** variables in a dimensionally homogeneous equation and if these contain **m** fundamental dimensions, then the variables are arranged into (n – m) dimensionless terms. These dimensionless terms are called Π terms.

3. What are all the advantages of dimensional analysis?

1. It expresses the functional relationship between the variables in dimensional terms.
2. It enables getting up a theoretical solution in a simplified dimensionless form.
3. The results of one series of tests can be applied to a large number of other similar problems with the help of dimensional analysis.

4. What are all the limitations of dimensional analysis?

1. The complete information is not provided by dimensional analysis. It only indicates that there is some relationship between the parameters.
2. No information is given about the internal mechanism of physical phenomenon.
3. Dimensional analysis does not give any clue regarding the selection of variables.

5. Define Reynolds number (Re).

It is defined as the ratio of inertia force to viscous force.

$$Re = \frac{\text{Inertiaforce}}{\text{Viscousforce}}$$

6. Define Prandtl number (Pr).

It is the ratio of the momentum diffusivity to the thermal diffusivity.

$$Pr = \frac{\text{Momentumdiffusivity}}{\text{Thermal diffusivity}}$$

7. Define Nusselt Number (Nu).

It is defined as the ratio of the heat flow by convection process under an unit temperature gradient to the heat flow rate by conduction under an unit temperature gradient through a stationary thickness (L) of metre.

$$\text{Nusselt Number (Nu)} = \frac{q_{conv}}{q_{cond}}$$

8. Define Grash of number (Gr).

It is defined as the ratio of product of inertia force and buoyancy force to the square of viscous force.

$$Gr = \frac{\text{Inertiaforce} \times \text{Buoyancyforce}}{(\text{Viscousforce})^2}$$

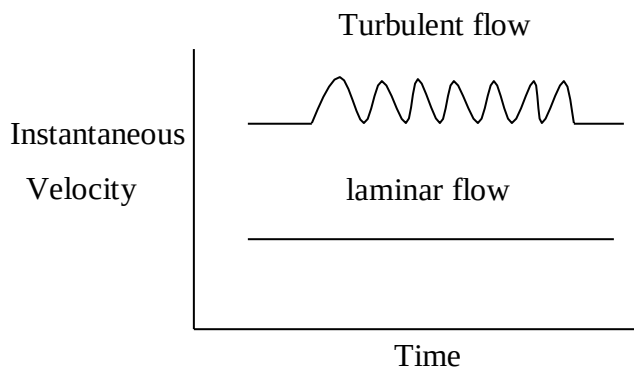
9. Define Stanton number (St).

It is the ratio of Nusselt number to the product of Reynolds number and Prandtl number.

$$St = \frac{Nu}{Re \times Pr}$$

10. What is meant by laminar flow and turbulent flow?

Laminar flow: Laminar flow is sometimes called stream line flow. In this type of flow, the fluid moves in layers and each fluid particle follows a smooth continuous path. The fluid particles in each layer remain in an orderly sequence without mixing with each other.



Turbulent flow: In addition to the laminar type of flow, a distinct irregular flow is frequently observed in nature. This type of flow is called turbulent flow. The path of any individual particle is zig - zag and irregular. Fig. shows the instantaneous velocity in laminar and turbulent flow.

11. Define convection.

Convection is a process of heat transfer that will occur between a solid surface and a fluid medium when they are at different temperatures.

12. State Newton's of convection.

Heat transfer from the moving fluid to solid surface is given by the equation

$$Q = hA(T_w - T_\infty)$$

This equation is referred to as Newton's law of cooling.

Where

h – Local heat transfer coefficient in $\text{W/m}^2 \text{K}$

A – Surface area in m^2

T_w – Surface (or) Wall temperature in K

T_∞ – Temperature of fluid in K

13. What is meant by free or natural convection?

If the fluid motion is produced due to change in density resulting from temperature gradients, the mode of heat transfer is said to be free or natural convection.

14. What is forced convection?

If the fluid motion is artificially created by means of an external force like a blower or fan, that type of heat transfer is known as forced convection.

15. What is the form of equation used to calculate heat transfer for flow through cylindrical pipes?

$$\text{Nu} = 0.023 (\text{Re})^{0.8} (\text{Pr})^n$$

$n = 0.4$ for heating of fluids.

$n = 0.3$ for cooling of fluids.

16. What are the dimensionless parameters used in forced convection?

1. Reynolds number (Re).

2. Nusselt number (Nu).

3. Prandtl number (Pr).

17. Define boundary layer thickness.

The thickness of the boundary layer has been defined as the distance from the surface at which the local velocity or temperature reaches 99% of the external velocity or temperature.

18. Indicate the concept or significance of boundary layer.

In the boundary layer concept the flow field over a body is divided into two regions:

- A thin region near the body called the boundary layer where the velocity and the temperature gradients are large.
- The region outside the boundary layer where the velocity and the temperature gradients are very nearly equal to their free stream values.

19. Write down the momentum equation for a steady, two dimensional flow of an incompressible, constant property Newtonian fluid in the rectangular coordinate system and mention the physical significance of each term.

Momentum equation,

$$P \left[u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] = F_x - \frac{\partial P}{\partial x} + \mu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]$$

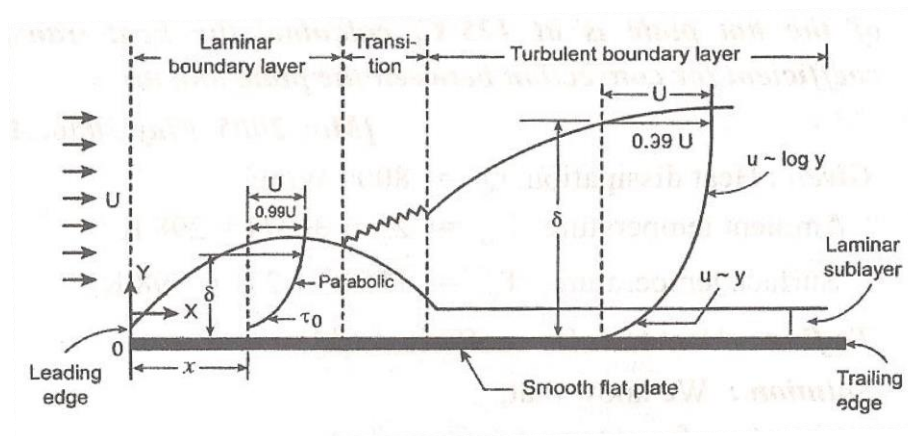
where, $P \left[u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] = \text{Inertia forces.}$

$F_x = \text{Body force.}$

$\frac{\partial P}{\partial x} = \text{Pressure force.}$

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \text{Viscous forces.}$

20. Sketch the boundary development of a flow.



21. Define displacement thickness.

The displacement thickness is the distance, measured perpendicular to the boundary, by which the free stream is displaced on account of formation of boundary layer.

22. Define momentum thickness.

The momentum thickness is defined as the distance through which the total loss of momentum per second be equal to if it were passing a stationary plate.

23. Define energy thickness.

The energy thickness can be defined as the distance, measured perpendicular to the boundary of the solid body, by which the boundary should be displaced to compensate for the reduction in kinetic energy of the flowing fluid on account of boundary layer formation.

24. Give the application of boiling and condensation.

Boiling and condensation process finds wide applications as mentioned below

1. Thermal and nuclear power plant
2. Refrigerating systems
3. Process of heating and cooling
4. Air conditioning systems

25. What is meant by pool boiling?

If heat is added to a liquid from a submerged solid surface, the boiling process is referred to as pool boiling. In this case the liquid above the hot surface is essentially stagnant and its motion near the surface is due to free convection and mixing induced by bubble growth and detachment.

26. What is meant by Filmwise condensation?

The liquid condensate wets the solid surface, spreads out and forms a continuous film over the entire surface is known as film wise condensation.

27. What is meant by Dropwise condensation?

In dropwise condensation, the vapour condenses into small liquid droplets of various sizes which fall down the surface in a random fashion.

28. Give the merits of dropwise condensation.

In dropwise condensation, a large portion of the area of the plate is directly exposed to vapour. The heat transfer rate in dropwise condensation is 10 times higher than in film condensation.

29. Write the force balance equation on a volume element for filmwise condensation on a vertical plane surface.

$$\frac{\partial^2 U}{\partial y^2} = \frac{1}{\mu_l} \frac{\partial p}{\partial x} - \frac{Bx}{\mu_l}$$

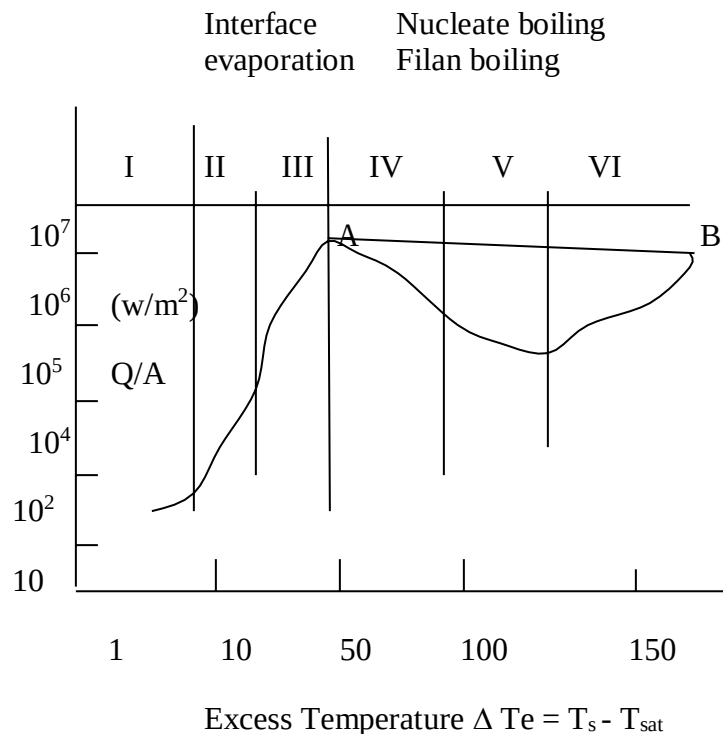
Where,

Bx – Body force in x direction

$\frac{\partial p}{\partial x}$ – Pressure gradient

30. Draw different regions of boiling and what is nucleate boiling?

Nucleate boiling exists in regions II and III. The nucleate boiling begins at region II. As the excess temperature is further increased, bubbles are formed more rapidly and rapid evaporation takes place. This is indicated in region III. Nucleate boiling exists up to $\Delta T = 50^\circ\text{C}$.



I – Free convection

II – Bubbles condense in super heated liquid

III – Bubbles raise to surface

IV – Unstable film

V – Stable film VI – Radiation coming into play

31. What are the types of heat exchangers?

The types of heat exchangers are as follows

- Direct contact heat exchangers
- Indirect contact heat exchangers
- Surface heat exchangers
- Parallel flow heat exchangers
- Counter flow heat exchangers
- Cross flow heat exchangers
- Shell and tube heat exchangers
- Compact heat exchangers

UNIT-III

HEAT TRANSFER BY RADIATION

1. Define Radiation.

The heat transfer from one body to another without any transmitting medium is known as radiation. It is an electromagnetic wave phenomenon.

2. Define emissive power [E_b].

The emissive power is defined as the total amount of radiation emitted by a body per unit time and unit area. It is expressed in W/m^2 .

3. Define monochromatic emissive power. [E_{b2}].

The energy emitted by the surface at a given length per unit time per unit area in all directions is known as monochromatic emissive power.

4. What is meant by absorptive?

Absorptive is defined as the ratio between radiation absorbed and incident radiation.

$$\text{Absorptivity, } \alpha = \frac{\text{Radiation absorbed}}{\text{Incident radiation}}$$

5. What is meant by reflectivity?

Reflectivity is defined as the ratio of radiation reflected to the incident radiation.

$$\text{Reflectivity, } \rho = \frac{\text{Radiation reflected}}{\text{Incident radiation}}$$

6. What is meant by transmissivity?

Transmissivity is defined as the ratio of radiation transmitted to the incident radiation.

$$\text{Transmissivity, } \tau = \frac{\text{Radiation transmitted}}{\text{Incident radiation}}$$

7. What is black body?

Black body is an ideal surface having the following properties.

1. A black body absorbs all incident radiation, regardless of wave length and direction.
2. For a prescribed temperature and wave length, no surface can emit more energy than black body.

8. State Planck's distribution law.

The relationship between the monochromatic emissive power of a black body and wave length of a radiation at a particular temperature is given by the following expression, by Planck.

$$E_b \lambda = \frac{c_1 \lambda^{-5}}{\left[\left(\frac{c_2}{\lambda T} \right) - 1 \right]}$$

$$E_{b\lambda} = \text{Monochromatic emissive power } W / m^2$$

$$\lambda = \text{Wavelength} - m$$

Where

$$c_1 = 0.374 \times 10^{-15} W m^2$$

$$c_2 = 14.4 \times 10^{-3} m k$$

9. State Wien's displacement law.

The Wien's law gives the relationship, between temperature and wave length corresponding to the maximum spectral emissive power of the black body at that temperature.

$$\lambda_{\max} T = c_3$$

Where →

$$c_3 = 2.9 \times 10^{-3}$$

$$\lambda_{\max} T = 2.9 \times 10^{-3} m k [\text{Radiation constant}]$$

10. State Stefan-Boltzmann law.

The emissive power of a black body is proportional to the fourth power of absolute temperature.

$$E_b \propto T^4$$

$$E_b = \sigma T^4$$

$$\text{Where } E_b = \text{Emissive power, } W/m^2$$

$$\sigma = \text{Stefan-Boltzmann constant}$$

$$= 5.67 \times 10^{-8} W/m^2 K^4$$

$$T = \text{Temperature, } k$$

11. Define Emissivity.

It is defined as the ability of the surface of a body to radiate heat. It is also defined as the ratio of emissive power of anybody to the emissive power of a black body of equal temperature.

$$\text{Emissivity, } \epsilon = \frac{E}{E_b}$$

12. What is meant by gray body?

If a body absorbs a definite percentage of incident radiation irrespective of their wavelength, the body is known as gray body. The emissive power of a gray body is always less than that of the black body.

13. State Kirchhoff's law of radiation.

This law states that the ratio of total emissive power to the absorptivity is constant for all surfaces which are in thermal equilibrium with the surroundings. This can be written as

$$\frac{E_1}{\alpha_1} = \frac{E_2}{\alpha_2} = \frac{E_3}{\alpha_3}$$

It also states that the emissivity of the body is always equal to its absorptivity when the body remains in thermal equilibrium with its surroundings.

$$\alpha_1 = \epsilon_1; \alpha_2 = \epsilon_2 \quad \text{and so on}$$

14. Define intensity of radiation (I_b).

It is defined as the rate of energy leaving a space in a given direction per unit solid angle per unit area of the emitting surface normal to the mean direction in space.

$$I_n = \frac{E_b}{\Pi}$$

15. State Lambert's cosine law.

It states that the total emissive power E_b from a radiating plane surface in any direction is proportional to the cosine of the angle of emission.

$$E_b \propto \cos \theta$$

16. What is the purpose of radiation shield?

Radiation shields constructed from low emissivity (high reflective) materials. It is used to reduce the net radiation transfer between two surfaces.

17. Define irradiation (G) and radiosity (J)

It is defined as the total radiation incident upon a surface per unit time per unit area. It is expressed in W/m^2 .

It is used to indicate the total radiation leaving a surface per unit time per unit area. It is expressed in W/m^2 .

18. What are the assumptions made to calculate radiation exchange between the surfaces?

- All surfaces are considered to be either black or gray.
- Radiation and reflection process are assumed to be diffuse.
- The absorptivity of a surface is taken equal to its emissivity and independent of temperature of the source of the incident radiation.

19. What is meant by shape factor and mention its physical significance.

The shape factor is defined as “The fraction of the radiative energy that is diffused from one surface element and strikes the other surface directly with no intervening reflections”. It is represented by F_{ij} . Other names for radiation shape factor are view factor, angle factor and configuration factor. The shape factor is used in the analysis of radiative heat exchange between two surfaces.

20. Discuss the radiation characteristics of carbon dioxide and water vapour.

The CO_2 and H_2O both absorb and emit radiation over certain wavelength regions called absorption bands.

The radiation in these is a volume phenomenon.

The emissivity of CO_2 and the emissivity of H_2O at a particular temperature increases with partial pressure and mean beam length.

UNIT-IV

HEAT EXCHANGER

1. What is heat exchanger?

A heat exchanger is defined as an equipment which transfers the heat from a hot fluid to a cold fluid.

2. What are the types of heat exchangers?

The types of heat exchangers are as follows

1. Direct contact heat exchangers
2. Indirect contact heat exchangers
3. Surface heat exchangers
4. Parallel flow heat exchangers
5. Counter flow heat exchangers
6. Cross flow heat exchangers
7. Shell and tube heat exchangers
8. Compact heat exchangers

3. What is meant by direct heat exchanger (or) open heat exchanger?

In direct contact heat exchanger, the heat exchange takes place by direct mixing of hot and cold fluids.

4. What is meant by indirect contact heat exchanger?

In this type of heat exchangers, the transfer of heat between two fluids could be carried out by transmission through a wall which separates the two fluids.

5. What is meant by Regenerators?

In this type of heat exchangers, hot and cold fluids flow alternately through the same space.

Examples: IC engine, gas turbines.

6. What is meant by Recuperators (or) Surface heat exchangers?

This is the most common type of heat exchangers in which the hot and cold fluid do not come into direct contact with each other but are separated by a tube wall or a surface.

Examples: Automobile radiators, Air preheaters, Economisers etc.

7. What is meant by parallel flow heat exchanger?

In this type of heat exchanger, hot and cold fluids move in the same direction.

8. What is meant by counter flow heat exchanger?

In this type of heat exchanger, hot and cold fluids move in parallel but opposite directions.

9. What is meant by cross flow heat exchanger?

In this type of heat exchanger, hot and cold fluids move at right angles to each other.

10. What is meant by Shell and tube heat exchanger?

In this type of heat exchanger, one of the fluids move through a bundle of tubes enclosed by a shell. The other fluid is forced through the shell and it moves over the outside surface of the tubes.

11. What is meant by compact heat exchangers?

There are many special purpose heat exchangers called compact heat exchangers. They are generally employed when convective heat transfer co-efficient associated with one of the fluids is much smaller than that associated with the other fluid.

12. What is meant by LMTD?

We know that the temperature difference between the hot and cold fluids in the heat exchanger varies from point to point. In addition various modes of heat transfer are involved. Therefore based on concept of appropriate mean temperature difference, also called logarithmic mean temperature difference, the total heat transfer rate in the heat exchanger is expressed as

$$Q = UA(\Delta T)_m$$

Where

U = Overall heat transfer co-efficient, W/m²K

A = Area, m²

$(\Delta T)_m$ = Logarithmic mean temperature difference.

13. What is meant by Fouling factor?

We know the surface of a heat exchangers do not remain clean after it has been in use for some time. The surfaces become fouled with scaling or deposits. The effect of these deposits

affecting the value of overall heat transfer co-efficient. This effect is taken care of by introducing an additional thermal resistance called the fouling resistance.

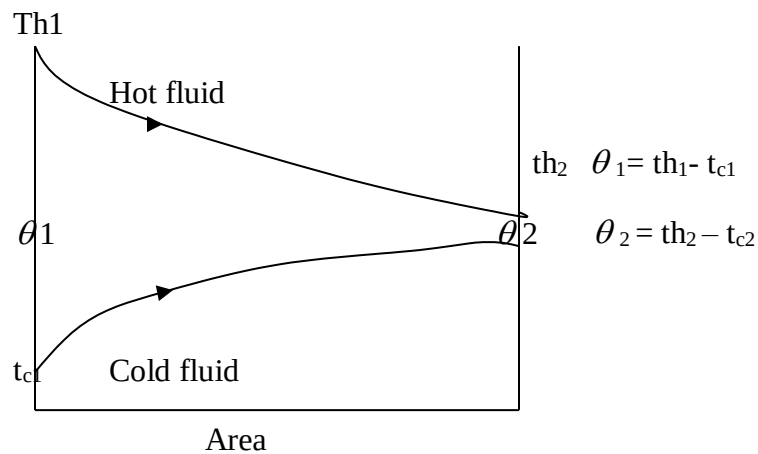
14. What is meant by Effectiveness?

The heat exchanger effectiveness is defined as the ratio of actual heat transfer to the maximum possible heat transfer.

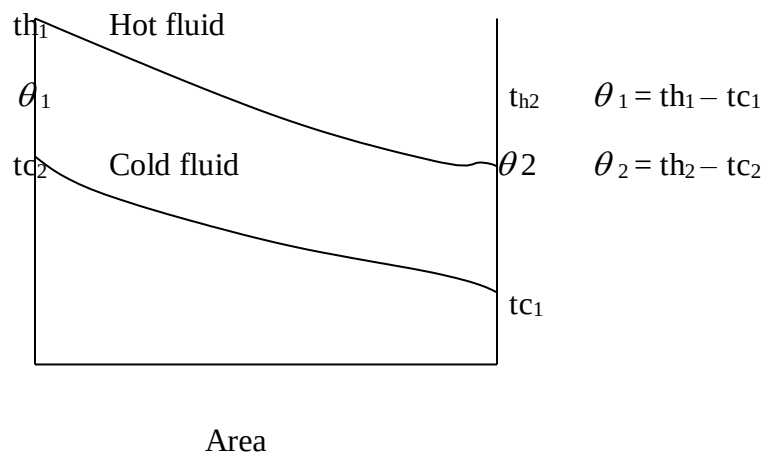
$$\text{Effectiveness } \epsilon = \frac{\text{Actual heat transfer}}{\text{Maximum possible heat transfer}}$$

$$= \frac{Q}{Q_{\max}}$$

15. Sketch the temperature variations in parallel flow and counter flow heat exchangers.



Temperature distribution – Parallel flow



Temperature distribution – Counter flow

UNIT-V

MASS TRANSFER

1. What is mass transfer?

The process of transfer of mass as a result of the species concentration difference in a mixture is known as mass transfer.

2. Give the examples of mass transfer.

Some examples of mass transfer are

- Humidification of air in cooling tower
- Evaporation of petrol in the carburetor of an IC engine
- The transfer of water vapour into dry air

3. What are the modes of mass transfer?

There are basically two modes of mass transfer,

- Diffusion mass transfer
- Convective mass transfer

4. What is molecular diffusion?

The transport of water on a microscopic level as a result of diffusion from a region of higher concentration to a region of lower concentration in a mixture of liquids or gases is known as molecular diffusion.

5. What is Eddy diffusion?

When one of the diffusion fluids is in turbulent motion, eddy diffusion takes place.

6. What is convective mass transfer?

Convective mass transfer is a process of mass transfer that will occur between a surface and a fluid medium when they are at different concentrations.

7. State Fick's law of diffusion.

The diffusion rate is given by the Fick's law, which states that molar flux of an element per unit area is directly proportional to concentration gradient.

$$\frac{m_a}{A} = -D_{ab} \frac{dC_a}{dx}$$

Where

$$\frac{m_a}{A} = -\text{molarflux} - \frac{\text{kg} - \text{mole}}{\text{s} - \text{m}^2}$$

$$D_{ab} - \text{Diffusioncoefficient of species a and b, m}^2 / \text{s}$$

$$\frac{dC_a}{dx} - \text{Concentration gradient, kg / m}^3$$

8. What is free convective mass transfer?

If the fluid motion is produced due to change in density resulting from concentration gradients, the mode of mass transfer is said to be free or natural convective mass transfer.

Example: Evaporation of alcohol.

9. Define forced convective mass transfer.

If the fluid motion is artificially created by means of an external force like a blower or fan, that type of mass transfer is known as convective mass transfer.

Example: The evaporation of water from an ocean when air blows over it.

10. Define Schmidt Number.

It is defined as the ratio of the molecular diffusivity of momentum to the molecular diffusivity of mass.

$$Sc = \frac{\text{Molecular diffusivity of momentum}}{\text{Molecular diffusivity of mass}}$$

11. Define Sherwood Number.

It is defined as the ratio of concentration gradients at the boundary.

$$Sh = \frac{h_m x}{D_{ab}}$$

$$h_m - \text{Mass transfer coefficient, m / s}$$

$$D_{ab} - \text{Diffusion coefficient, m}^2 / \text{s}$$

$$x - \text{Length, m}$$

12. Give two examples of convective mass transfer.

- Evaporation of alcohol
- Evaporation of water from an ocean when air blows over it.

13. Define the following.

- Mass Concentration
- Molar Concentration
- Mass fraction
- Mole fraction